- The method is simple to use.
- It is to assess sampling error in the method.

Demerits:

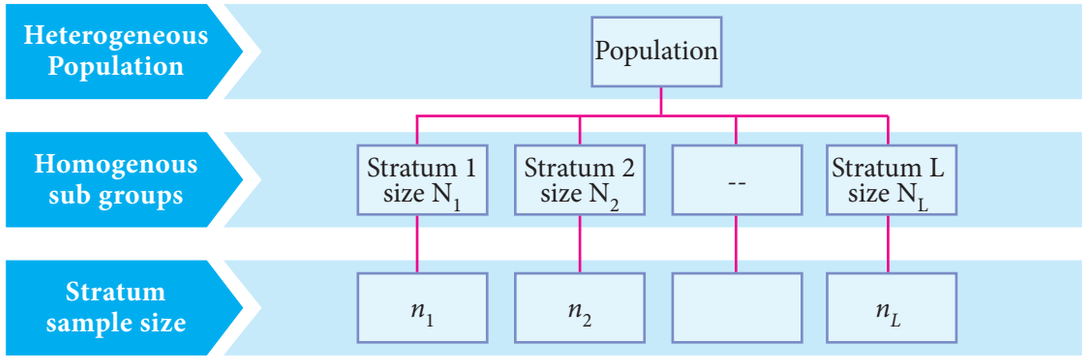
- If the units are widely dispersed, the sample becomes unrepresentative.
- The method is not applicable when the units are heterogeneous in nature.
- Units selected may be too widely dispersed; thus, adherence to the whole sample may be difficult. Remark: Simple random sampling is more suitable when the population is homogeneous. If the population is heterogeneous we must go in for stratified random sampling.

NOTE

Random number can also be generated through scientific calculator or through the computer.

2.7.2 Stratified Random Sampling

In Stratified random sampling, the heterogeneous population of size N units is sub-divided into L homogeneous non overlapping sub populations called Strata, the ith stratum having N_i units (i =1, 2, 3,…,L) such that $N_1+N_2+\ldots+N_L = N$.



A sample size being n_i from i^{th} stratum (i=1, 2,…,L) is independently taken by simple random sampling in such way that $n_1+n_2+\ldots+n_L = n$. A sample obtained using this procedure is called a stratified random sample.

To determine the sample size for each stratum, there are two methods namely proportionate allocation and optimum allocation. In proportionate allocation sample size is determined as proportionate to stratum size. If the stratum size large, that stratum will get more representation in the sample. If the stratum size small, that stratum will get less representation in the sample. The sample size for the ith stratum can be determined using the formula $n_i = (n/N)^* N_i$. The optimum allocation method uses variation in the stratum and cost to determine the stratum sample size ni .

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